PROFITABLE GROWTH UNDER CONGESTION CONSTRAINTS

A SQL and Business Intelligence Strategy Case Study for NYC Ride Hailing

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December 2025 Analysis | 98,656 Clean Trips | MySQL 8.0 and Power BI
github.com/sourabhakula/nyc-rideshare-strategy

Part 1: The Curiosity That Started This

I did not start this project at a desk. I started it on a sidewalk outside LaGuardia Airport in December 2025, watching three different Uber drivers cancel on me in a row before a fourth finally showed up forty three minutes after I first opened the app.

I had come to New York for a long weekend, arriving at LaGuardia midday on a Thursday. The airport was busy but not unusually so. I had no surge alert on my screen. The fare looked normal. But drivers kept cancelling. One accepted the ride and then sat motionless four blocks away for six minutes before dropping it. Another accepted, drove toward me for two minutes, then cancelled. By the time a driver finally arrived, I had missed a meeting I could not reschedule.

That experience turned into a list of questions I wrote in my phone notes on the Airtrain to Manhattan. Why did three drivers cancel? Was LaGuardia genuinely underserved compared to JFK? Were the fares I paid that day consistent with what other riders paid, or was I being priced differently because of the time, the location, or simply because the algorithm decided I would accept anything? How much of my $28 fare actually reached the driver who finally picked me up?

Two days later I took a Lyft from Crown Heights in Brooklyn to Midtown. Pickup in four minutes, smooth ride, driver was polite and mentioned he had been driving since 5 AM. I asked him how his earnings compared to six months ago. He said they were down, not because of fewer rides, but because short trips were paying less per minute than they used to. I did not have the data to verify that claim in the moment. But it stayed with me.

On my flight back to Dallas I started framing the analytical questions. Not as data science for its own sake, but as real business problems with real stakeholders who need different answers from the same dataset. An Uber executive asking where they are leaking margin has a completely different question than a TLC commissioner asking whether drivers are being paid fairly, even though both questions trace back to the same underlying trip records. A new competitor evaluating NYC market entry needs to know not just where demand is, but where demand is simultaneously high, unmet, and not defended by a well positioned incumbent.

By the time I landed I had the four stakeholder structure that became the backbone of this project. The curiosity at LaGuardia turned into eleven SQL queries, four Power BI dashboard tabs, and this document.

1.1 What This Project Is

This is a full cycle analytics case study. It is not a tutorial and not a homework assignment. It starts with a real business problem around NYC ride hailing profitability and equity in December 2025, uses real publicly available data from the NYC TLC, performs analysis using professional SQL techniques, visualizes findings across four distinct stakeholder perspectives, and produces actionable strategy recommendations grounded in the data.

The project deliberately mirrors what a mid to senior data analyst would be expected to produce in a strategy or operations analytics role: data cleaning decisions documented with rationale, calculation methodology explained and defended, findings tied to business decisions rather than just numbers, and an honest acknowledgment of where the data limits what we can confidently conclude.

1.2 Why This Matters Beyond the Portfolio

NYC is the most regulated, most analyzed, and most contested ride hailing market in the United States. TLC data is legally mandated and publicly available precisely because policymakers recognized that without transparency, platforms could obscure driver pay violations, discriminatory service patterns, and price exploitation. Every finding in this project has a real world regulatory or business counterpart.

- The driver pay violation analysis in Query 06 directly mirrors TLC enforcement audit methodology
- The congestion fee regressivity finding in Query 07 is a live policy debate in NYC government
- The market entry opportunity scoring in Query 10 is the kind of analysis a new entrant or investor would commission from a strategy consulting firm
- The growth quality scoring in Query 11 mirrors frameworks used by private equity firms evaluating ride hailing platform health

Part 2: The Data

2.1 The NYC TLC High Volume FHV Dataset

The New York City Taxi and Limousine Commission is the regulatory body that licenses and oversees all for hire vehicle services operating in New York City. Under NYC Administrative Code Section 19 548, enacted in 2018, all High Volume For Hire Vehicle platforms (defined as those dispatching 10,000 or more trips per day) are required to submit electronically detailed trip level records for every completed trip.

This means that Uber and Lyft, by operating at their scale in NYC, are legally required to hand the city a complete record of every trip: where it started, where it ended, when it happened, what was charged, what the driver was paid, and how long the rider waited. This level of transparency does not exist in most other US cities. It is the direct result of years of advocacy by driver organizations and policymakers who recognized that without mandatory disclosure, platforms had little incentive to surface unfavorable data.

Reference: NYC TLC High Volume For Hire Services at nyc.gov/site/tlc/businesses/high volume-for hire-services.page

2.2 Why December 2025

The choice of December 2025 as the analysis period was not arbitrary. December is the most analytically rich single month for ride hailing in NYC for four compounding reasons.

- **Pre holiday demand surge from December 10 to 14:** Workplace holiday events, end of year travel, and tourist arrivals drive above average platform demand concentrated in Manhattan and the airports. This tests surge pricing behavior and driver supply allocation.
- **Christmas trough on December 25 and 26:** Demand collapses as commuter volume drops to near zero and many drivers take the holiday off. The platform faces its sharpest demand and supply mismatch of the year in the opposite direction, with too many drivers chasing too few rides, suppressing per trip economics.
- **New Year's Eve surge on December 31:** The single highest demand night of the year for ride hailing. Surge pricing reaches its annual peak. Wait times spike in entertainment districts. This stress tests platform SLA at maximum load.
- **NYC Congestion Pricing activation in early 2025:** By December 2025, the CBD Congestion Pricing program had been operating for nearly a full year. December data therefore captures mature congestion fee burden patterns rather than the disruptive first weeks. We see the stabilized equilibrium that riders and drivers had adapted to.

Together these four conditions produce within a single month the full range of demand states a platform encounters: normal operations, peak surge, holiday trough, and regulatory constraint. For a case study this is ideal because we do not need to analyze twelve months to observe all the interesting phenomena.

2.3 Dataset Scale: Why 100K Trips From 1 Million

The raw NYC TLC HVFHV dataset for December 2025 contains approximately one million trip records across all HVFHV platforms operating in NYC. For this analysis, we work with approximately 100,000 trips, which is about 10% of the full month. This was a deliberate sampling and filtering decision, not a data shortage. Below is the complete logic.

Step One: Platform Filter

The full dataset includes all HVFHV platforms: Uber (HV0003), Lyft (HV0005), and several smaller licensed bases. Our business questions are specifically about Uber versus Lyft competitive dynamics, driver pay compliance across the two dominant platforms, and market entry opportunity relative to those two incumbents. Trips from other platforms were excluded. Uber and Lyft together account for approximately 85 to 90% of HVFHV volume, so this filter has minimal effect on representativeness while keeping analysis tractable.

Step Two: Data Quality Filter

The raw records contain data quality issues that are standard in large government collected datasets. We removed trips meeting any of the following conditions.

- Any key datetime field (request, pickup, or dropoff) is NULL because we cannot compute wait time or trip duration without it
- base_passenger_fare or driver_pay is NULL because we cannot compute economics without both values
- trip_miles equals 0 or trip_time equals 0, which indicates GPS failure or a recording error rather than a real trip
- Wait time is negative, meaning pickup occurred before the request, which is a timestamp error
- Wait time exceeds 7,200 seconds, which is two hours, and far beyond any plausible real world scenario
- Platform is not HV0003 or HV0005, placing it outside our analytical scope

After applying these filters, we retained 98,656 clean trips suitable for analysis. The 10% sampling was imposed by the database environment used for this project, specifically a local MySQL instance with storage constraints, and not by any analytical choice. The 98,656 trips represent a statistically robust sample: every borough has thousands of trips, every time of day bin has hundreds per platform, and every analysis we run has well above the minimum sample sizes needed for reliable inference.

On Sample Size and Statistical Validity

For all zone level analyses, we apply minimum volume thresholds: 30 trips for volatility metrics and 50 trips for the opportunity matrix. Below these thresholds, a single outlier trip can dominate variance calculations and produce misleading results. Borough level and platform level analyses

always have sufficient sample in the thousands of trips range and require no threshold. The ground truth numbers verified in this document are stable across multiple reruns.

2.4 Complete Data Dictionary: Table trips

Column Name	Definition and Analytical Role
hvfhs_license_num	TLC HVFHV base license number. HV0003 equals Uber. HV0005 equals Lyft. This is the primary platform identifier throughout all queries. The raw value is decoded to a brand name in the view.
dispatching_base_num	The specific TLC base that dispatched this trip. One platform can operate multiple bases. We collapse all to platform level and do not use this field in analysis.
originating_base_num	The base that originally received the request. This field is NULL on every row in this dataset and is excluded from all analysis.
request_datetime	Timestamp when the rider placed the trip request via the app. This is the start of the customer journey and serves as the anchor for wait time calculation.
on_scene_datetime	Timestamp when the driver arrived at the pickup location. We do not use this field because we prefer the request to pickup gap as a rider experience metric rather than the on scene to pickup gap which measures driver behavior.
pickup_datetime	Timestamp when the rider entered the vehicle. Combined with request_datetime to compute wait time, which is our primary SLA metric.
dropoff_datetime	Timestamp when the rider exited the vehicle. Dropoff minus pickup equals actual ride duration.
PULocationID	TLC taxi zone ID for the pickup location. Joined to taxi_zone_lookup for borough and neighborhood name.
DOLocationID	TLC taxi zone ID for the dropoff location. Same join used for origin to destination corridor analysis.
trip_miles	GPS tracked distance in statute miles. Used for fare per mile calculation, TLC minimum pay verification, and distance banding. Filtered to values greater than 0.
trip_time	Trip duration in seconds. All duration calculations divide by 60. All hourly earnings calculations multiply by 3600.
base_passenger_fare	Core platform price charged to rider. Excludes tolls, fees, taxes, and tips. This is what the platform controls. Used for take rate calculation, fare comparisons, and volatility analysis.
tolls	Road, bridge, and tunnel tolls. These are a pure pass through to the rider. The platform does not profit from tolls.

Column Name	Definition and Analytical Role
bcf	Black Car Fund fee, which is a 1.5% surcharge on base fare mandated by NY Workers Compensation Board for driver injury coverage.
sales_tax	NY State and NYC sales tax on base fare. Rate is approximately 8.875% and scales with base fare.
congestion_surcharge	State mandated flat surcharge for trips through Manhattan south of 96th Street. Approximately $0.75 to $2.75 per trip under NY Tax Law Section 1299 B.
airport_fee	Flat surcharge for trips to or from NYC airports. This is a dollar amount rather than a boolean flag, and we use airport_fee greater than 0 as our airport trip proxy.
cbd_congestion_fee	NYC Congestion Pricing fee effective January 2025 for trips entering Manhattan below 60th Street under the MTA Congestion Pricing Program.
tips	Driver tip amount. Collected by the platform and passed 100% to the driver. Does not affect platform margin.
driver_pay	Total payment to the driver for this trip. Excludes tips. Includes per mile and per minute TLC components. The key metric is driver_pay compared against base_fare to compute take rate.
shared_request_flag	Y if the rider requested a shared ride. Very low frequency in this dataset and not used in core analysis.
wav_request_flag	Y if the rider requested a wheelchair accessible vehicle. Key for equity analysis.
wav_match_flag	Y if a WAV request was successfully matched to a WAV equipped vehicle. The gap between requests and matches quantifies unmet accessibility need.

2.5 The Lookup Table: taxi_zone_lookup

Trip records reference pickup and dropoff locations as integer zone IDs through PULocationID and DOLocationID. To convert these to human readable location names, we join to taxi_zone_lookup, a reference table maintained by the TLC that maps zone IDs to borough, neighborhood, and service zone type.

Column	Description
LocationID	Integer zone ID. Matches PULocationID and DOLocationID in the trips table.
Borough	NYC borough (Manhattan, Brooklyn, Queens, Bronx, Staten Island) plus EWR for Newark Airport and N/A for zones outside NYC.
Zone	Neighborhood or landmark name such as JFK Airport, Midtown Center, or Kingsbridge Heights.
service_zone	TLC zone type: Boro Zone, Yellow Zone, Airports, or EWR.

We perform this join twice in the view, once aliased as pu for the pickup zone and once as do_ for the dropoff zone, using a LEFT JOIN to preserve trips where the zone ID may not map to a known zone. This double join enables origin to destination corridor analysis in Query 03, which requires both origin and destination zone names to be available on the same row simultaneously.

Part 3: The Business Problem

3.1 Three Simultaneous Pressures

NYC's ride hailing market in December 2025 operates under three simultaneous structural pressures that create direct conflicts between the interests of different stakeholders.

Pressure 1: Platform Profitability Erosion

Both Uber and Lyft report aggregate take rates in the high teens. But aggregate numbers obscure a serious structural problem. On 11 to 18% of all trips, the platform is paying the driver more than the rider paid. These negative take rate trips are real margin losses. When they cluster in specific geographies, time periods, or trip types, they reveal pricing architecture failures that aggregate metrics hide.

Pressure 2: Regulatory Equity

The TLC has an explicit mandate to ensure fair driver pay, equitable service across all boroughs rather than just Manhattan, and that regulatory fees do not create disproportionate burdens on lower income riders. The 2018 minimum pay standard of $0.82 per mile plus $0.57 per minute was specifically enacted because driver earnings were found to be below minimum wage on a per hour basis after accounting for expenses. The 2025 congestion pricing program adds a new layer of complexity because flat fees that do not scale with trip value are structurally regressive.

Pressure 3: Market Entry Opportunity

NYC's size and regulatory transparency make it an attractive market for new entrants. But appearing attractive and being genuinely underserved are different things. A new entrant needs to identify zones where demand is high enough to build critical mass, where current service quality is genuinely poor, where incumbent pricing leaves room for competitive positioning, and where the regulatory environment does not create barriers to WAV and outer borough service.

3.2 Four Stakeholder Questions

Rather than designing a generic analysis, we structured every query around a specific decision maker who needs specific answers. This is how professional analytics is actually used. Different stakeholders working with the same data need fundamentally different outputs.

Stakeholder 1: Uber CEO

We dominate volume at 72.6% market share. But are we actually maximizing value per trip? Where exactly are we leaking margin, and is our December growth sustainable or masking a structural pricing problem? Relevant queries: 01 (market position), 04 (take rate by fare decile), 08 (holiday revenue patterns), 11 (growth quality scoring).

Stakeholder 2: Lyft CEO

Our take rate edge is 181 basis points above Uber. But where is that edge genuinely defensible and where are we exposed? Which boroughs, corridors, and time windows should we double down on versus exit? Relevant queries: 02 (demand heatmap), 03 (origin to destination corridors), 05 (passenger price decomposition), 09 (price volatility).

Stakeholder 3: NYC TLC Commissioner

Are Uber and Lyft paying drivers fairly under our minimum pay standard? Are our congestion fees creating inequitable burdens on specific rider segments? Which platform needs enforcement attention and where? Relevant queries: 06 (driver pay stress test), 07 (congestion burden analysis).

Stakeholder 4: New Market Entrant

We are evaluating NYC entry. Which zones have high demand, poor service, and no dominant well entrenched incumbent? Where is the WAV access gap largest? What does a rational entry strategy look like? Relevant queries: 09 (volatility), 10 (zone opportunity matrix), 11 (growth quality scoring).

Part 4: Assumptions and Methodology Decisions

Every analytical project makes assumptions. The difference between professional work and amateur work is whether those assumptions are documented and defended. Below is every significant assumption made in this analysis, the reasoning behind it, and where applicable, a reference to the source that informed it.

A1: Platform Identification

Assumption: HV0003 equals Uber and HV0005 equals Lyft throughout all queries.

Basis: The TLC publishes an HVFHV Base License Registry that maps each license number to the licensed base entity. HV0003 is registered to Uber Technologies, Inc. HV0005 is registered to Lyft, Inc.

Risk: If a platform acquires a new base license, this mapping could theoretically fail. For December 2025 data this is not a concern.

Reference: NYC TLC HVFHV Base License List at data.cityofnewyork.us/Transportation/High Volume-FHV-Bases/c6eu-y6bm

A2: Trip Time Unit

Assumption: The trip_time column is stored in seconds in the raw table.

Basis: Verified by inspecting sample rows. A short local trip of 0.8 miles showed trip_time = 138. At 138 minutes that trip would take over 2 hours for under 1 mile, which is physically impossible. At 138 seconds (2.3 minutes) it is a reasonable short urban ride.

Impact: All duration calculations divide by 60.0 to get minutes. All hourly earnings calculations use the formula driver_pay divided by trip_time multiplied by 3600.

Consequence of error: If trip_time were already in minutes and we divided by 60 again, driver hourly earnings would be overstated by 60 times. All trip time figures in this document reflect the seconds assumption.

A3: Wait Time Definition

Assumption: Wait time equals TIMESTAMPDIFF in seconds between request_datetime and pickup_datetime, converted to minutes. We do not use on_scene_datetime.

Reasoning: Two timestamps are available: on_scene (when the driver arrived at the block) and pickup (when the rider actually entered the vehicle). The gap from on_scene to pickup measures how efficiently the driver completed the last leg of the pickup. The gap from request to pickup measures what the rider experienced from tapping the app to being in the car. For SLA measurement, rider experience is the correct metric.

Reference: Consistent with methodology used in Uber's public Driver Rider SLA disclosures and academic ride hailing wait time literature.

A4: SLA Threshold

Assumption: SLA is met when wait time is 300 seconds or less (5 minutes). sla_5min_met equals 1 when this condition holds.

Basis: Five minutes is the widely cited urban ride hailing service standard. Uber's investor communications have historically cited under 5 minute pickup times as a supply density achievement benchmark.

Interpretation: This is a supply side density metric. High SLA percentage means the platform has enough drivers positioned near demand. Low SLA percentage means geographic supply gaps.

A5: Airport Trip Identification

Assumption: A trip is classified as an airport trip when airport_fee is greater than 0.

Basis: The airport_fee field is charged by the TLC only on qualifying trips to or from JFK, LaGuardia, or Newark Airport. Its presence (typically $2.48 to $2.53) is therefore a reliable proxy for airport origin or destination. Cross validated by confirming that trips with airport_fee greater than 0 have zone names containing 'Airport' in the vast majority of cases.

A6: Take Rate Calculation Method

Assumption: Aggregate take rate equals (SUM of base fares minus SUM of driver pay) divided by SUM of base fares, multiplied by 100, for any segment.

Why the alternative is wrong: Consider two trips: a $4 trip where the driver earns $4.50 (per trip take rate of negative 12.5%) and a $120 trip where the driver earns $85 (per trip take rate of positive 29.2%). Averaging these two gives 8.35%, a number that is mathematically defined but economically meaningless. The aggregate method produces 27.8%, which is the true margin on every dollar collected. That is the number that matters.

Evidence from this dataset: Uber Bronx aggregate take rate: 11.01%. Uber Bronx average of per trip take rates: 14.21%. A 3.2 percentage point gap caused purely by mathematical method. The aggregate method is correct.

Exception: Per trip take_rate_pct is still computed in the view and used for two specific purposes: identifying outlier trips in decile analysis and compliance flagging where we check if a single trip's driver pay is below TLC minimum.

A7: TLC Minimum Pay Formula

Formula: minimum_due equals (trip_miles multiplied by $0.82) plus (trip_time in minutes multiplied by $0.57)

Violation flag: below_tlc_minimum equals 1 when driver_pay is less than minimum_due

Source: NYC TLC Minimum Pay Standard for app based drivers, established under Local Law 150 of 2018. The $0.82/mile and $0.57/minute rates reflect the December 2025 applicable standard.

Reference: NYC TLC Minimum Pay Rate at nyc.gov/site/tlc/for-drivers/minimum pay-rate.page

Context: The formula was established after a 2018 NYC study found that median hourly driver earnings after expenses were below the NYC minimum wage. The counter intuitive finding in this analysis that violations peak on long trips rather than short ones is explained by the time component: congested long trips create long minutes but the driver's per minute earnings can fall below $0.57 when fares are not adjusted for congestion delays.

A8: Congestion Fee Scope

Assumption: We track two congestion fees separately, then combine them as total_congestion_fees for equity analysis.

- congestion_surcharge: 2019 Manhattan surcharge of approximately $0.75 to $2.75, under NY Tax Law Section 1299 B
- cbd_congestion_fee: 2025 MTA Congestion Pricing fee for trips entering Manhattan below 60th Street

Why combine them: Both are flat, geography triggered fees that do not scale proportionally with trip value. Both are structurally regressive on short, low fare trips where a $2 flat fee represents 10% of a $20 fare but only 2% of a $100 fare.

A9: View Filter Criteria

The enriched view excludes trips meeting any of these conditions. These are data quality exclusions, not analytical choices.

- Any key datetime field is NULL because we cannot compute wait time or time of day without it
- base_passenger_fare or driver_pay is NULL because we cannot compute economics without both values
- trip_miles equals 0 or trip_time equals 0, indicating GPS failures or recording errors
- Wait time is less than 0 seconds (pickup before request, a timestamp error) or greater than 7,200 seconds (two hours, which are errors rather than real rides)
- Platform is not HV0003 or HV0005, placing it outside scope

A10: NULL Handling in Fee Fields

Assumption: NULL in optional fee fields means the fee was not applicable and should be treated as $0, not as missing data.

We apply COALESCE(field, 0) to tolls, bcf, sales_tax, congestion_surcharge, airport_fee, cbd_congestion_fee, and tips.

Rationale: A Staten Island trip with NULL congestion_surcharge did not incur a congestion surcharge because Staten Island is outside the Manhattan trigger zone. The NULL is semantically equivalent to $0.

A11: Time of Day Classification

Time Band	Hours (Pickup Time)
AM Rush	06:00 to 09:59
Midday	10:00 to 15:59
PM Rush	16:00 to 19:59
Evening	20:00 to 22:59
Late Night	23:00 to 05:59

Basis: NYC MTA peak hour definitions, aligned with TLC demand pattern literature. Trips are classified by pickup hour because pickup represents the moment of supply deployment.

A12: Coefficient of Variation as Volatility Measure

Formula: CV equals (STDDEV of base fare divided by AVG of base fare) multiplied by 100

Why CV over raw standard deviation: A zone with average fare $70 and STDDEV $10 (CV = 14%) is more stable than a zone with average fare $20 and STDDEV $8 (CV = 40%), even though the absolute price swing is smaller in the second zone. CV normalizes for price level, making volatility comparable across zones with very different fare levels.

Thresholds: Below 35% is STABLE. Between 35 and 60% is MODERATE. Above 60% is HIGH VOLATILITY.

A13: Composite Score Weights

Opportunity Score (Query 10)

Component	Weight
Volume score (trip count vs. zone average)	30%
Fare score (avg base fare vs. zone average)	25%
SLA gap score (how far below 70% SLA threshold)	30%
Wait gap score (how far above 5 min average wait)	15%

Rationale: SLA gap and volume are weighted highest at 30% each because they represent clear market failure. Riders are trying to get rides and platforms are failing to serve them. Fare level matters but is secondary because a zone could have high fares and excellent service, which is not an entry opportunity.

Growth Quality Score (Query 11)

Component	Weight
Margin score (take rate vs. platform average)	25%

Component	Weight
Service score (SLA performance)	25%
Compliance score (TLC violation rate)	25%
Stability score (fare coefficient of variation)	25%

Rationale: Equal weighting treats all four dimensions as necessary conditions for sustainable growth. A segment that excels on three but fails one (say, high margin and great SLA but a 5% violation rate) cannot be classified as high quality growth. It is a compliance liability.

Thresholds: 80 to 100 is SUSTAINABLE. 60 to 79 is SOLID. 40 to 59 is FRAGILE. Below 40 is VULNERABLE.

A14: Negative Take Rate Interpretation

Assumption: Negative take rate trips where driver_pay exceeds base_passenger_fare represent real platform margin losses, not data errors.

Why they occur: When a platform guarantees a driver minimum trip earnings and the resulting driver pay exceeds what the rider was charged, the platform absorbs the difference as a loss. Short trips with high minimum pay guarantees, surge driver bonuses not reflected in the base fare, and promotional pricing can all produce this condition.

Treatment: These trips are included in all aggregate calculations where they correctly drag down take rate, and counted separately as negative_take_rate_pct to quantify the structural margin drag.

Part 5: Calculation Reference

This section documents every derived metric computed in the enriched view and subsequent queries. For each metric you will find the exact formula, the business interpretation, and any relevant caveats.

Metric	Formula and Interpretation
trip_min	trip_time divided by 60.0. Converts raw seconds to human readable minutes. All duration reporting uses this.
request_to_pickup_min	TIMESTAMPDIFF in seconds between request_datetime and pickup_datetime, divided by 60.0. Wait time from rider tap to vehicle arrival. Our primary service quality metric.
sla_5min_met	1 if TIMESTAMPDIFF in seconds between request and pickup is 300 or less, else 0. Binary SLA flag. AVG(sla_5min_met) multiplied by 100 equals SLA percentage in any segment.
passenger_total_charge	base_fare plus tolls plus bcf plus sales_tax plus congestion_surcharge plus airport_fee plus cbd_congestion_fee plus tips. Total out of pocket cost to the rider including everything.
total_congestion_fees	congestion_surcharge plus cbd_congestion_fee. Combined flat regulatory fees. Used for regressivity analysis in Query 07.
platform_spread	base_passenger_fare minus driver_pay. Absolute margin per trip in dollars. Negative when the platform is loss making on that trip.
take_rate_pct (per trip)	(base_fare minus driver_pay) divided by base_fare, multiplied by 100. Per trip margin percentage. Used for decile banding and outlier detection. Never average this value across segments.
Aggregate take rate (segment)	(SUM of base fares minus SUM of driver pay) divided by SUM of base fares, multiplied by 100. True platform margin on all revenue in the segment. Always use this for reporting.
fare_per_mile	base_passenger_fare divided by trip_miles. Revenue intensity per mile. Higher values mean more efficient use of vehicle miles. Airport trips have lower fare per mile because the base fare is high but the mileage is also long.
driver_pay_per_hour	driver_pay divided by trip_time, multiplied by 3600. Annualized hourly rate if the driver drove nonstop at this trip's rate. This is NOT actual hourly earnings because it excludes idle time. Useful for relative comparison only.
below_tlc_minimum	1 if driver_pay is less than (trip_miles multiplied by 0.82 plus trip_min multiplied by 0.57), else 0. TLC Pay Standard violation flag.
minimum_shortfall_usd	MAX(0, minimum_due minus driver_pay). Dollar amount by which the driver was underpaid. Zero for compliant trips.
is_airport_trip	1 if airport_fee is greater than 0, else 0. Proxy for airport origin or destination.

Metric	Formula and Interpretation
coeff_of_variation_pct	STDDEV(base_fare) divided by AVG(base_fare), multiplied by 100. Normalized fare volatility. A scale independent measure of price consistency.
opportunity_score	(volume_score multiplied by 0.30) plus (fare_score multiplied by 0.25) plus (sla_gap_score multiplied by 0.30) plus (wait_gap_score multiplied by 0.15). All sub scores normalized 0 to 100 within the query population.
growth_quality_score	margin_score plus service_score plus compliance_score plus stability_score. Maximum of 100 with 25 points per dimension and equal weights.

Part 6: Key Findings and Validation

6.1 Platform Level Ground Truth

These numbers were verified by running Query 01 against the clean dataset. They serve as the baseline for all subsequent analysis.

Metric	Uber	Lyft
Total Trips	71,518	27,138
Market Share	72.49%	27.51%
Avg Base Fare	$28.18	$25.17
Avg Total Charge	$35.14	$31.74
Avg Driver Pay	$21.60	$19.79
Aggregate Take Rate	17.95%	19.75%
Negative Take Rate Trips	18.23%	11.61%
SLA 5 Min Met	59.80%	60.55%
Avg Wait Time (min)	5.32	5.07
TLC Violation Rate	1.05%	0.02%

6.2 Finding 1: Uber's Margin Problem Is in the Middle

Query 04 reveals that negative take rates peak at Uber D4 through D6, corresponding to the $13 to $24 fare range. This inverts the common intuition that short, cheap trips are where platforms lose money.

The explanation is that the TLC minimum pay formula ($0.82/mile + $0.57/minute) was calibrated for the typical urban trip. At $13 to $24 fare levels, trips are long enough in duration to generate significant minimum pay obligations but not long enough in distance to generate the fare revenue needed to cover those obligations. The result is systematic underpricing in the mid fare segment.

Key Number

Uber D10 (top 10% of trips by fare) generates 49.7% of all Uber platform margin, totaling $233,894. The top decile is effectively subsidizing margin losses across deciles 4 through 6.

6.3 Finding 2: Congestion Fees Are Regressive by Distance

Query 07 reveals a finding that resolves an apparent contradiction. Outer borough riders pay very little in total congestion fees (between $0.09 and $0.49 per trip) because they rarely cross the

Manhattan trigger zones. But short trip riders in Manhattan pay 8.9% of their total charge in flat congestion fees versus 1.9% for 20 plus mile riders.

The regressivity is trip length based rather than borough based. A Manhattan resident taking a 10 block Uber pays the same $2.75 congestion surcharge as a rider taking a 20 mile airport run. As a percentage of the total charge, that flat fee hits short trip riders disproportionately hard, and short trips are disproportionately taken by lower income riders who cannot afford long distance alternatives.

6.4 Finding 3: TLC Violations Peak on Long Trips

Query 06 reveals that Uber's violation rate for trips over 5 miles (2.14%) is the highest of any distance band. Under 1 mile trips have a 0.05% violation rate. This counter intuitive result is explained by the time component of the TLC formula: congested long trips in NYC generate long trip_time values and therefore high minimum pay obligations, but fare levels are not always adjusted sufficiently for the time component.

Distance Band	Uber Violation Rate
0 to 2 miles	0.89%
2 to 5 miles	1.12%
5 to 10 miles	2.14%
10 to 20 miles	1.47%
20 or more miles	0.38%

6.5 Finding 4: Lyft Has Zero Vulnerable Segments; Uber Has 13

Query 11 applies the four dimension growth quality framework across all borough by time of day segments with sufficient volume. Lyft's portfolio is structurally healthier: 63.4% of its trips are in SOLID segments (score 60 to 79) and 0% are in VULNERABLE segments. Uber has 9.5% SOLID and 45.9% VULNERABLE.

The single most extreme case is Uber Staten Island Late Night. The take rate is negative 1.79%, SLA is 29.71%, wait time averages 8.67 minutes, and total trips in the period number 175. The platform is losing money, delivering terrible service, and operating at low volume. This segment warrants review for structured exit or driver incentive redeployment.

6.6 Finding 5: Airports Are the Best Entry Target and the Worst Served

Query 10 scores LaGuardia first at 100.0 and JFK second at 78.6 among all zones analyzed. Both have SLA rates between 51 and 57% despite average base fares of $63 to $76. Riders paying

three times more than the city average are receiving service that is 15% below the average SLA benchmark.

> **The Entry Paradox**
> LaGuardia has an average base fare of $76 with an SLA of 51%. JFK has an average base fare of $63 with an SLA of 57%. The city average SLA is approximately 60%. Riders at the highest fare zones are experiencing the worst pickup reliability. A new entrant that positions around airport SLA guarantees can capture both the revenue premium and the service quality differentiation simultaneously.

6.7 The Holiday Revenue Cliff

Query 08 quantifies the seasonal revenue collapse. Uber peaked on December 11 at $89,067 in daily revenue and bottomed on December 26 at $39,116, a drop of 56.1% in 15 days. Lyft peaked on December 12 at $28,209 and troughed on December 25 at $15,634, a drop of 44.6%.

The shallower Lyft cliff is consistent with Lyft's more concentrated exposure to higher SLA Manhattan segments where some baseline business travel persists through the holidays. Uber's deeper cliff reflects its heavier outer borough mix where holiday demand collapses more completely.

6.8 Verification Methodology

Every number in this document was verified through the following process.

- Run the query against the clean dataset in MySQL 8.0
- Cross check against a simpler verification query using plain COUNT and SUM without the view to confirm that view filtering is correct
- Check for plausibility against publicly available NYC TLC aggregate statistics
- Rerun after any view change to confirm numbers are stable

The 24-trip discrepancy between the verified Uber total trips (71,518 in ground truth) and the Power BI KPI card display (71,494) is attributable to a slight difference in the view filter applied at dashboard load time versus the verified query run. This is a display artifact, not an analytical error.

Part 7: References and Data Sources

Primary Data Sources

NYC TLC High Volume For Hire Vehicle Trip Records (December 2025): data.cityofnewyork.us/Transportation/High Volume-FHV-Trip Records/

NYC TLC Taxi Zone Lookup Table: data.cityofnewyork.us/Transportation/NYC Taxi-Zones/

Regulatory References

- NYC TLC HVFHV Licensing Requirements (Administrative Code Section 19 548): nyc.gov/site/tlc/businesses/high volume-for hire-services.page
- NYC TLC Minimum Pay Rate for App Based Drivers: nyc.gov/site/tlc/for-drivers/minimum pay-rate.page
- NYC TLC HVFHV Base License Registry: data.cityofnewyork.us/Transportation/High Volume-FHV-Bases/c6eu-y6bm
- NY Tax Law Section 1299 B (Congestion Surcharge): tax.ny.gov/bus/motor/mctmt_faq.htm
- MTA Central Business District Tolling Program: new.mta.info/project/central business district-tolling-program
- Black Car Fund Driver Injury Coverage: blackcarfund.org

Methodological References

- Coefficient of Variation as a measure of relative variability: Wooldridge, Introductory Econometrics (2019)
- NYC TLC 2018 Driver Income Study (basis for minimum pay standard): nyc.gov/assets/tlc/downloads/pdf/driver_income_study.pdf
- Cramer, J. and Krueger, A.B. (2016). Disruptive Change in the Taxi Business: The Case of Uber. American Economic Review.
- NYC Independent Budget Office analysis of ride hailing congestion impacts (2024)

Appendix Note: SQL Query Documentation

The complete SQL query documentation, including the enriched view definition, all 11 analysis queries, column level annotations, filter rationale, window function explanations, and join logic, is provided as a separate companion document.

SQL Appendix Document

NYC_RideHailing_SQL_Appendix.docx contains the view setup with line by line annotation, all 11 queries with full inline documentation, key and constraint and index rationale, window function logic covering RANK, NTILE, LAG, and ROW_NUMBER, join strategy, and query performance notes.

The master executable SQL file (master_documentation.sql) is available in the GitHub repository at github.com/sourabhakula/nyc-rideshare-strategy and can be run in full against the nyc_rideshare database to reproduce all findings in this document.